2026Year 2nd Semester Syllabus

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Course Title	DIGITAL MICROELECTRONICS	Course Code-Section	EEE3511-02
Credit/Time/ Experiment,Lab,Practical Technique Time	3/Tue8,Thu7,8	Department	Electrical and Electronics Engineering
Time	Tue8,Thu7,8	Location	EngHD603
Exam Date & Time	Midterm exam	Final exam	
Class Language	English	Evaluation Type	Absolute evaluation

Instructor's Profile	Name	Jeong Hanwool	Contact Information	Telephone	
	Department	전기전자공학부		Mail	HANWOOL87@YONSEI.AC.KR
	Office			Interview information	

TA's Name & Contact Information	Name		Contact Information	Telephone	

Course Description Brief Introduction of the Course	This course introduces the basic principles and design methods of digital integrated circuits, following a structure largely based on the Docceptor Digital Integrated Circuit Design curriculum. Topics include MOSFET fundamentals, CMOS logic, delay and power, arithmetic circuits, memory circuits, and basic AI accelerator design. Through hands-on projects using TCAD, SPICE, layout verification, and RTL-to-GDS tools, students will experience the overall IC design flow and learn to complete simple designs independently. The projects focus more on practical tool experience than on design complexity. 본 교과목은 Digital Integrated Circuit Design 교육과정을 중심으로 디지털 집적회로의 기본 원리와 설계 방법을 다룹니다. MOSFET 기초, CMOS 논리회로, 지연 및 전력, 산술회로, 메모리 회로, AI 가속기 기초 등을 학습합니다. 또한 TCAD, SPICE, 레이아웃 검증 및 RTL-to-GDS 툴을 활용한 실습 프로젝트를 통해 집적회로 설계의 전체 흐름을 경험하고, 간단한 설계를 스스로 수행할 수 있는 능력을 기릅니다. 프로젝트는 설계 난이도보다 실제 툴 사용 경험에 중점을 둡니다.			
Course Goals	1.	Korean	Understand and analyze the basic structure and operating principles of digital integrated circuits.	40%
		English	Understand and analyze the basic structure and operating principles of digital integrated circuits.	
	2.	Korean	Interpret the characteristics of MOSFETs and CMOS logic gates and apply them to circuit design.	30%
		English	Interpret the characteristics of MOSFETs and CMOS logic gates and apply them to circuit design.	
	3.	Korean	Learn and utilize key design techniques such as RC delay modeling, power analysis, and layout fundamentals.	20%
		English	Learn and utilize key design techniques such as RC delay modeling, power analysis, and layout fundamentals.	

	4.	Korean	Develop comprehensive IC design skills through advanced logic, memory, arithmetic, and data conversion circuit design.	10%
		English	Develop comprehensive IC design skills through advanced logic, memory, arithmetic, and data conversion circuit design.	
	5.	Korean		0%
		English		

Core Competencies	The total measurable competencies must be 100%. Each course objective should set the competency as 25%. The core and major competencies should equal at least 50%.		
	Basic Academic Ability 50%	Methematical Thinking 30%	Problem-solving Ability 20%
Sub-Competencies/Learning Unit1			
Sub-Competencies/Learning Unit2			
Sub-Competencies/Learning Unit3			
Core Competencies(Liberal Arts)Major competency(	**Must reflect the interrelationship between core competencies (elective courses) and major competencies (major studies).**		
Basic Academic Ability	This course develops fundamental skills in analyzing and designing digital circuits based on MOSFET and CMOS principles. Through hands-on practice with TCAD, SPICE, layout verification, and RTL-to-GDS tools, students learn to apply theory to practical design and improve their circuit analysis, design-tool, problem-solving, and semiconductor system design skills. 본 교과목은 MOSFET 및 CMOS 회로의 기본 원리를 바탕으로 디지털 회로를 분석하고 설계하는 기초 능력을 기릅니다. TCAD, SPICE, 레이아웃 검증, RTL-to-GDS 실습을 통해 이론을 실제 설계에 적용하며, 회로 분석 능력, 설계 도구 활용 능력, 문제 해결 능력 및 반도체 시스템 설계 역량을 향상시킵니다.		
Sustainable Development Goals			
Average Recommended Amount of Learning per	Average Reading Volume		Average amount of writing(Based on A4)

Course Methods (%) Total Amount 100	Lecture	Practice Training	Presentation	Dabate	Team Project
	80%	10%	0%	10%	0%
Course Methods 2 Select Relevant Items	PBL Subject	Capstone Design	CBL, Social Innovation Course	Flipped Classroom	Work Experience,Internsh

Grading Policy(%) Total Amount 100 Free Input for Other Information	Midterm exam	Final exam	Quiz	Individual Assignment	Team Assignment	Attendance	Others
	30%	40%	0%	20%	0%	10%	0%

Assignment/ Report, Project Guide	Title of Assignment/Project Name, and Method of Filling Out	Submission Deadline	Type of Submission and Method
	Project-1 TCAD Simulation (DEVSIM)		
	Project-2 Custom Transistor-level Circuit Design & Layout (SPICE + LVS/DRC)		
	Project-3 AI Accelerator Pilot Digital Design (RTL-to-GDS, Synthesis-to-PnR,		

	OpenROAD based)		
Prerequisite	기초회로이론, 전자회로1, 디지털논리회로	Online Course Address	

Course Material	Course Material Name	Author	Publisher	Publish Year	ISBN
주교재	Integrated Circuit Design, 4/E	Weste	Pearson	2021	6100321696945

Main Learner Precautions	질문 받는 카톡 ID : hanwool87ca The projects will focus on hands-on practice. Students will learn how to use the tools and complete simple designs by themselves. The main goal is to gain practical experience and basic independent design skills, not to make complex designs. Therefore, the projects will be easier than in previous years. 프로젝트는 직접 설계하고 툴을 사용해보는 실습 경험에 중점을 둡니다. 복잡한 설계를 완성하는 것보다, 학생들이 각 설계 툴의 기본 흐름을 익히고 간단한 설계를 스스로 수행할 수 있는 능력을 갖추는 데 의의가 있습니다. 이에 따라 프로젝트의 설계 난이도는 예년(설계 자체가 매우 어려웠음)보다 낮게 구성할 예정입니다.
Attatchment	

Weekly Plan

week	Period	Weekly Topic & Contents	Remarks
1	2026-09-01 2026-09-07	MOSFET Theory	(9.1.) Fall semester classes begin (9.3. - 9.7.) Course add and drop period
2	2026-09-08 2026-09-14	MOSFET I-V, Capacitances Project-1 TCAD Simulation (DEVSIM) : MOSFET I-V, C-V Extraction	
3	2026-09-15 2026-09-21	CMOS Logic Gate	
4	2026-09-22 2026-09-28	RC Delay Model, Logical Effort	(9.24. - 9.25.) Chuseok Holidays 09.24 추석, 09.25 추석, 09.26 추석
5	2026-09-29 2026-10-05	Dynamic Power, Static Power, Layout basic	(10.5.) Alternative Holiday of the National foundation Day 10.03 개천절, 10.05 대체공휴일(개천절)
6	2026-10-06 2026-10-12	Swtich, Tri-state inverter	(10.7.) First third of the semester ends (10.9.) Hangul Proclamation Day 10.09 한글날
7	2026-10-13 2026-10-19	Pseudo nMOS, CVSL, Dynamic Logic	(10.13. - 10.15.) Course withdrawal period
8	2026-10-20 2026-10-26	Midterm	(10.20. - 10.26.) Midterm Examinations
9	2026-10-27 2026-11-02	Pass TR Logic, XOR-XNOR	(10.27. - 10.29.) Application Period for S/U evaluation
10	2026-11-03 2026-11-09	Adder Tree, Tree Adder Project-2 Custom Transistor-level Circuit Design & Layout (SPICE + LVS/DRC)	
11	2026-11-10 2026-11-16	Level Shifter	(11.13.) Second third of the semester ends

12	2026-11-17 2026-11-23	Latch and Flip-Flop	
13	2026-11-24 2026-11-30	Systolic Array & NPU basic Project-3 AI Accelerator Pilot Digital Design (RTL-to-GDS, Synthesis-to-PnR, OpenROAD based)	
14	2026-12-01 2026-12-07	Memory basic (SRAM)	
15	2026-12-08 2026-12-14	Memory basic (DRAM)	(12.8. - 12.14.) Self-study
16	2026-12-15 2026-12-21	Final	(12.15. - 12.21.) Final Examinations

Attendance Requirement

• Students who are absent for more than one-third of the total class hours will receive an F or NP grade regardless of examination results.
• Courses without midterm and/or final examinations will hold regular classes during the examination periods.

Support for Students with Disabilities(SWDs)

• SWDs can request accommodations related to lectures, assignments, or tests by contacting the course professor at the beginning of semester. However, accommodations may vary depending on the essentiality of lecture and discretion of professors.

[Lecture]
- Visual Impairment: alternative, braille, enlarged reading materials, note-taker
- Physical Impairment: alternative reading materials, access to classroom, note-taker, assigned seat
- Hearing Impairment: note-taker/stenographer, recording lecture
- Intellectual Disability/Autism: note-taker

[Assignments and Test]
- Visual/Physical/Hearing Impairment: (reasonable) extra days for submission, alternative type of assignment, extended test time, alternative type of test, arranging separate test room, and proctors, test ghostwriter
- Intellectual Disability/Autism: (reasonable) extra days for submission, alternative type of assignment

Safety Precautions

• Students required to complete laboratory safety training must finish the online training before the semester begins and submit the certificate at the first class session. Failure to do so may result in restricted class participation.
• Students in physical activity courses should complete appropriate warm-up exercises before class. Those with cardiovascular or chronic respiratory conditions are advised to consult a physician before participating.

Course Administration

• Notices regarding class cancellations, make-up classes, classroom changes, and other course-related matters may be posted on LearnUs. For updates, please check LearnUs regularly.